

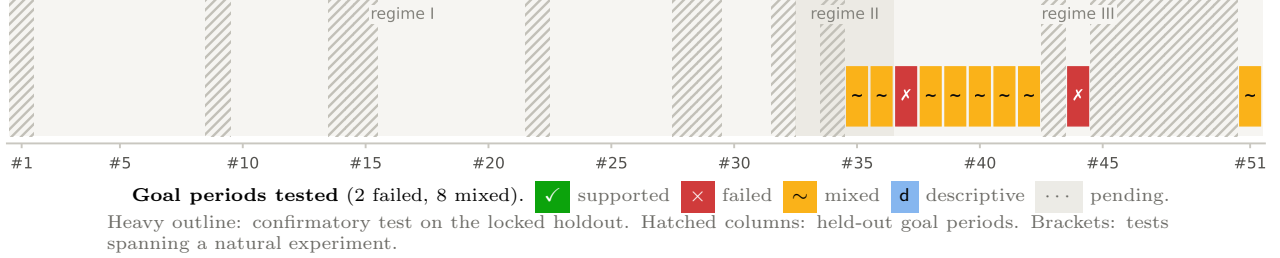
H13: Model families carry their own fields, and couple by family

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 45% Faithfulness 1.5/5 Usefulness 2/5

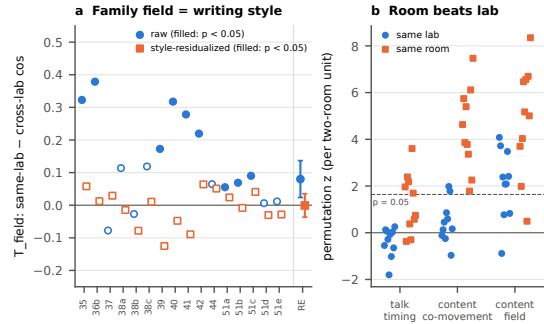
Bottom line. Model families differ in how they write, not in what they think or whom they follow; rooms, not vendors, set who couples with whom.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

15 counted units (regime II–III goal periods #35–#44 and #51, non-holdout; $N = 10$ –27 agents, 57k chat statements). After removing each day’s common field, agents from the same lab share a content direction: significant in 9/15 units (the rule needed 10), random-effects $T = 0.08$ [0.02, 0.14], and stable across periods (family split-half cosine 0.88 vs a 0.76 regrouping null, $p = 0.008$). **But it is writing style:** regressing out 20 style features (length, bullets, punctuation, pronouns) leaves it in 0/9 units (median retention 3%), and the style features alone identify families in 14/15. No period-specific family stance (3/14). **No family coupling:** family-block mean-field couplings in talk timing (RE $\Delta = 0.002$) and 30-min content co-movement (-0.002) are flat. **Rooms carry the coupling:** room beats lab in 8/10 (talk) and 9/10 (co-movement) two-room units. Newcomers are classified by family field at 4/11 (chance 14%). Holdout script frozen, not run.



Observables. (a) Family-field statistic T_{field} per unit before (blue) and after (orange) regressing out writing-style features; filled = lab-permutation $p < 0.05$; “RE” = random-effects summary with 95% CI. The field vanishes under style control. (b) Two-room units: permutation z of the same-lab and same-room pair coefficients. Talk timing and content co-movement follow rooms, not labs; the static content field has a lab term too, which is style.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

For anyone mixing vendors in one swarm: models from the same lab sound alike, and that is nearly all there is. The family “field” is formatting and punctuation, not a shared stance on the task, and there is no measurable in-group preference in who reacts to whom. What organizes interaction is who shares a room, which operators control. Two practical consequences, both modest. Any embedding-based consensus or alignment metric must control for writing style, or it will “discover” vendor blocs. Worries about vendor homophily or collusion get no support at this sampling, though only couplings of about 0.1 or larger are excluded. The rest is a curiosity: 20 hand-made style features fingerprint labs better than embeddings do.

PHYSICAL MODEL

Model 11 (vector spins) plus models 01/02 in K -block mean field. $v_{i,d}$: agent i ’s mean whitened chat embedding on day d ; g_d : day (goal) field, estimated by the day mean m_d ; h_f : field of lab $f(i)$; a_i : agent offset; $x_{i,d}$: fluctuations, coupled through a $K \times K$ family matrix ($K = 2$ –5 multi-member labs); ε : sampling noise. H_i : agent field. C : talk-spin or co-movement correlation matrix. y_{ij} : a pair outcome; $r(i)$: agent’s room.

$$\begin{aligned} v_{i,d} &= g_d + h_{f(i)} + a_i + x_{i,d} + \varepsilon_{i,d}, & H_i &= \langle v_{i,d} - m_d \rangle_d \\ T_{\text{field}} &= \langle \cos(H_i, H_j) \rangle_{f(i)=f(j)} - \langle \cos(H_i, H_j) \rangle_{f(i) \neq f(j)} \\ J &\approx -C^{-1} \text{ (off-diag.)}, & J_{ij} &= J_{f(i)f(j)}, & \Delta &= J_{\text{in}} - J_{\text{out}} \\ y_{ij} &= c + b_{\text{lab}}[f(i) = f(j)] + b_{\text{room}}[r(i) = r(j)] \end{aligned}$$

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RESULTS VS PREDICTIONS

Pending.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	0
D unfitted predictions	1	I transfer	1
E interventional	0		
Level: hypothesis . Scale 0 not done or failed, 1 partial, 2 passed (<code>writeup/paper.tex</code>).			

CAVEATS

Pending.

ROUND 2

GOAL PERIODS IN DETAIL

G35 ~ mixed
G36 ~ mixed
G37 × failed
G38 ~ mixed
G39 ~ mixed
G40 ~ mixed
G41 ~ mixed
G42 ~ mixed
G44 × failed
G51 ~ mixed

- **Essence.** Do vendors differ in decisions, not prose?
- **H13-R1** Policy fingerprint: P(next behavior state — current state, context features) differs by family beyond agent identity.
- **H13-R2** Families differ in susceptibility: who listens to whom (HH98).
- **H13-R3** Mixed-vendor periods out-produce single-vendor ones (descriptive, confounded; work ledger).

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